LABORATORY REPORT 9: STANDARD ACL CONFIGURATION

STUDENT INFORMATION

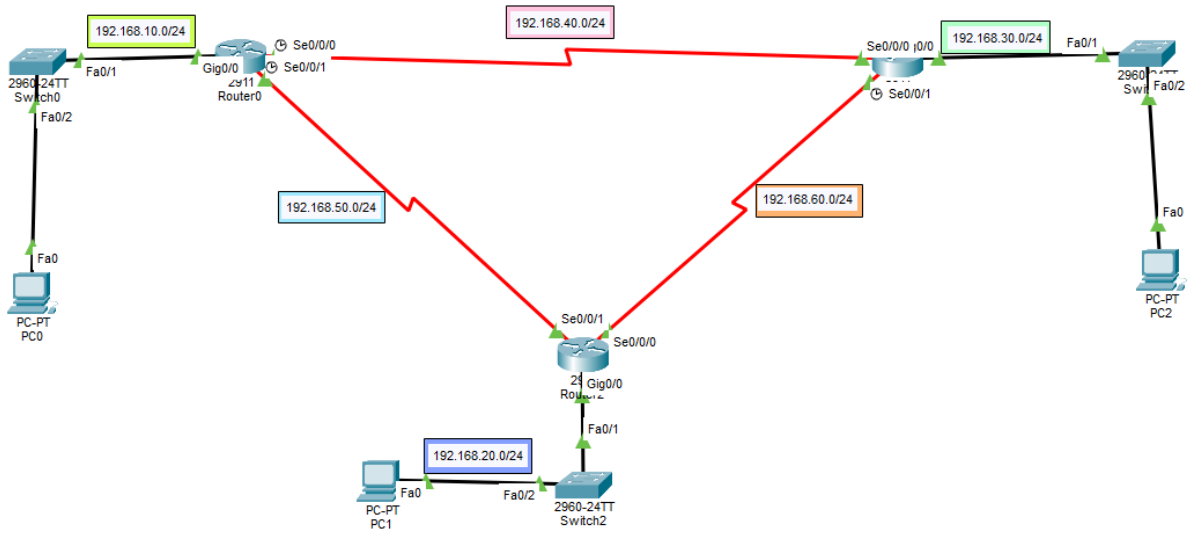
Item	Details
Student Name	Deepak Luitel
Enrollment Number	12250005
Date of Submission	17/05/2026

OBJECTIVE

To configure IP addresses on router interfaces, switches, and PCs to enable communication between devices on different networks using Standard Access Control Lists (ACLs) to filter traffic based on source IP addresses across multiple routers.

IP ADDRESS DERIVATION

For this practical, use the topology diagram from the previous practical, namely LABORATORY REPORT 8: DYNAMIC ROUTING (OSPF) CONFIGURATION, and continue with the same setup.



STANDARD ACL CONFIGURATION

Scenario 1: Block PC0 from accessing PC1 (Source-Based Filtering)

Objective: Prevent PC0 (192.168.X.2) from communicating with PC1 (192.168.(X+10).2) while allowing all other traffic.

ACL Placement: Standard ACLs should be placed as close to the destination as possible (on Router1's inbound interface or Router1's Gig0/0 outbound).

Configuration on Router1:

```
configure terminal

! Create Standard ACL 10 to deny PC0 and permit all others
access-list 10 deny host 192.168.5.2
access-list 10 permit any

! Apply ACL to inbound traffic on Serial0/0/0 (coming from Router0)
interface serial0/3/0
ip access-group 10 in
exit

! Alternative: Apply to outbound on Gig0/0 (towards PC1)
! interface gigabitethernet0/0
! ip access-group 10 out

end
write memory
```

Scenario 2: Allow only PC2 to access PC1 (Permit Specific, Deny All)

Objective: Allow only PC2 (192.168.(X+50).2) to communicate with PC1, block all others.

Configuration on Router1:

```
configure terminal

! Create Standard ACL 20 to permit only PC2
access-list 20 permit host 192.168.25.2

! Apply ACL to inbound traffic on Serial0/0/0
interface serial0/0/0
ip access-group 20 in
exit

end
```

```
write memory
```

Scenario 3: Block entire subnet from accessing PC2

Objective: Block entire PC0 subnet (192.168.X.0/24) from accessing PC2.

Configuration on Router2:

```
configure terminal

! Create Standard ACL 30 to deny PC0 subnet
access-list 30 deny 192.168.X.0 0.0.0.255
access-list 30 permit any

! Apply ACL to inbound on Serial0/0/0 (from Router0)
interface serial0/0/0
ip access-group 30 in
exit

end
write memory
```

VERIFICATION

```
show access-lists
show ip interface serial0/0/0
show running-config | include access-list
```

Screenshot 1: ACL Configuration
(Paste your ACL configuration screenshot here)

Screenshot 2: Access List Verification
(Paste your show access-lists output here)

CONNECTIVITY TESTING (WITHOUT ACL - BEFORE)

Test	Source	Destination	Command	Result
PC0 to PC1	PC0	PC1	ping 192.168.(X+10).2	Successful

PC2 to PC1	PC2	PC1	ping 192.168.(X+10).2	Successful
PC0 to PC2	PC0	PC2	ping 192.168.(X+50).2	Successful

CONNECTIVITY TESTING (WITH STANDARD ACL - AFTER)

Scenario 1 Results (Block PC0 from PC1):

Test	Source	Destination	Command	Expected Result	Actual Result
PC0 to PC1	PC0	PC1	ping 192.168.15.2	Failed	Failed
PC2 to PC1	PC2	PC1	ping 192.168.15.2	Successful	Successful
PC0 to PC2	PC0	PC2	ping 192.168.25.2	Successful	Successful

Scenario 2 Results (Allow only PC2 to PC1):

Test	Source	Destination	Command	Expected Result	Actual Result
PC0 to PC1	PC0	PC1	ping 192.168.15.2	Failed	Failed
PC2 to PC1	PC2	PC1	ping 192.168.15.2	Successful	Successful

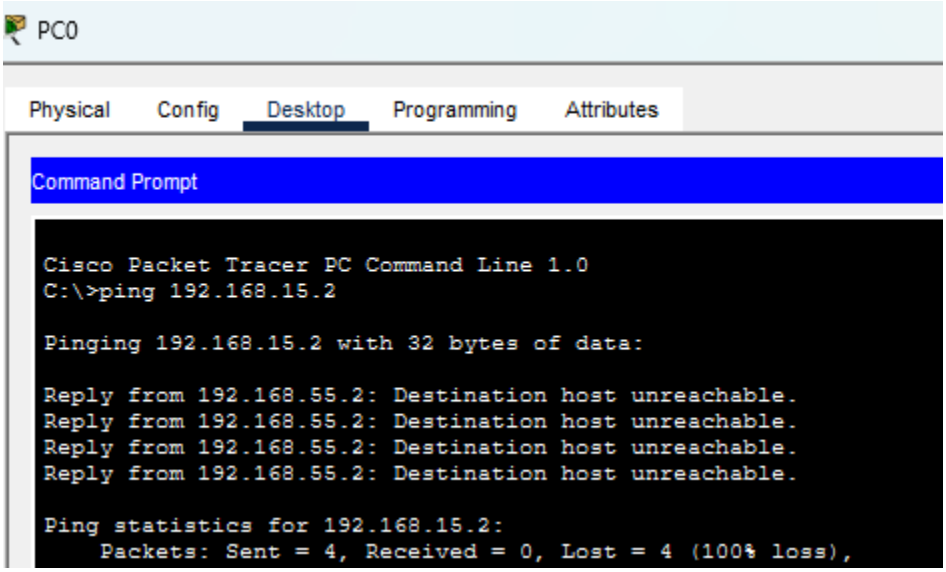
Scenario 3 Results (Block entire PC0 subnet from PC2):

Test	Source	Destination	Command	Expected Result	Actual Result
PC0 to PC2	PC0	PC2	ping 192.168.25.2	Failed	Failed
PC1 to PC2	PC1	PC2	ping 192.168.25.2	Successful	Successful

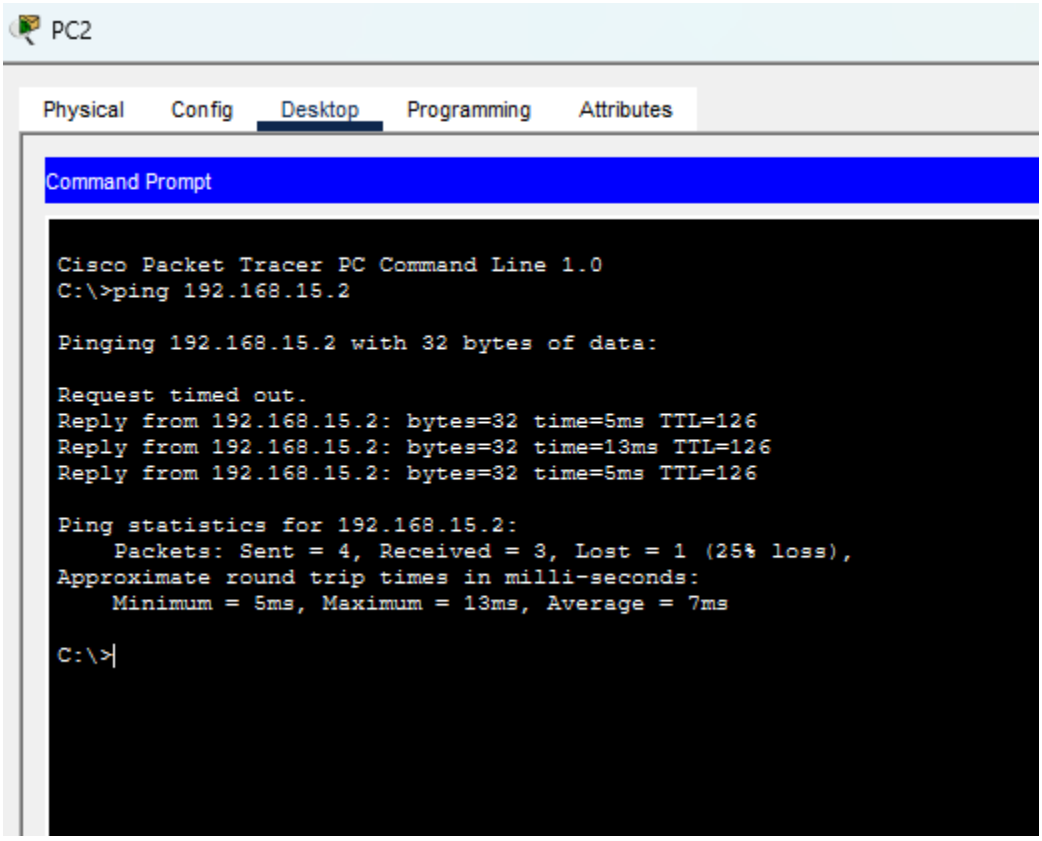
Screenshot 3: Ping Test Results (with ACL)

(Paste your ping test screenshot here)

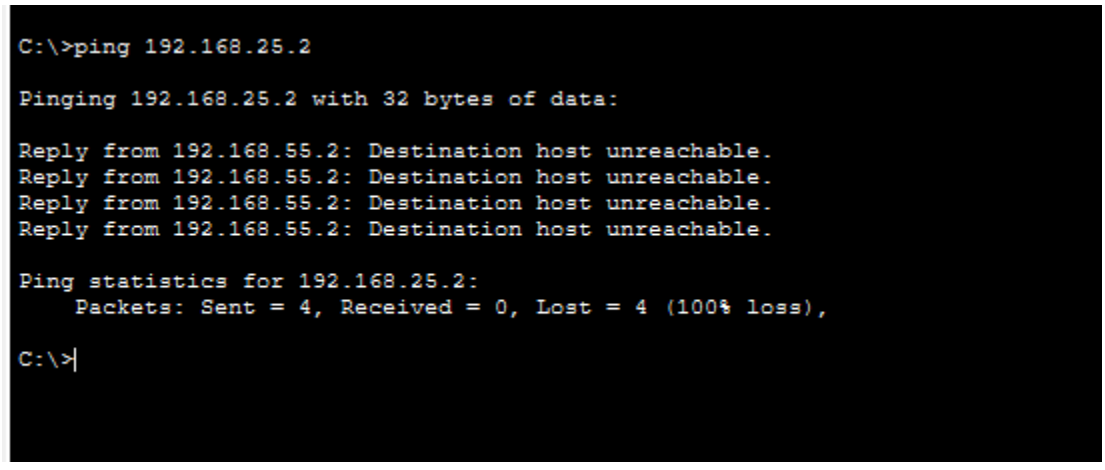
PC0 to PC1:



PC2 to PC1:



PC2 to PC1:



SHORT ANSWER QUESTIONS

1. **What is a Standard ACL and what does it filter based on?**

Filters traffic based on source IP address only, using rules numbered 1–99.

2. **Why should Standard ACLs be placed close to the destination?**

To avoid blocking the source from reaching other networks unintentionally.

3. **What is the difference between** `deny host 192.168.X.2` **and** `deny 192.168.X.0 0.0.0.255`**?**

First blocks one specific host, second blocks the entire subnet.

4. **What does the implicit deny any at the end of every ACL mean?**

Any packet not matching an ACL rule is automatically dropped.

5. **In Scenario 1, why did we place the ACL on Router1 instead of Router0?**

Router0 would block PC0 from all networks; Router1 only blocks PC0 from PC1.

CONCLUSION

Provide a concise summary (5–7 sentences):

Standard ACLs were configured on Cisco routers to filter traffic based on source IP addresses. Three scenarios were tested: blocking a single host, permitting only one host, and blocking an entire subnet. ACLs were placed close to the destination to avoid unintended traffic blocking. The implicit deny any ensured all unmatched traffic was dropped automatically. Verification commands confirmed the ACLs worked as expected.

DECLARATION

I hereby declare that I have performed this laboratory work and all submitted evidence represents my own practical implementation.

Student Signature: _____D.L_____

Date: _____17/05/2026_____